

MindEase – Data Collection & Sensor Integration Report

Application: MindEase – Stress Detection Android App

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Author: Technical Documentation

1. Heart Rate Measurement (Automatic Input)

Technology Used

Photoplethysmography (PPG) — the same principle used by hospital pulse oximeters and smartwatches.

How It Works

1. The user opens the camera scan screen and selects **"Measure with Camera"**
2. The **rear camera** activates with the **flashlight ON**
3. The user places their **finger firmly over the camera lens**
4. The flashlight illuminates the fingertip — blood flowing through capillaries causes **tiny brightness variations** in the camera image with each heartbeat
5. The app captures camera frames at ~30 FPS for **20 seconds**

Algorithm (Zero-Crossing Peak Detection)

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Step 1: Collect raw pixel brightness values from each frame
Step 2: Smooth the signal using a moving average (window size = 5)
Step 3: Calculate the mean of the smoothed signal
Step 4: Count "zero-crossings" – each time the signal crosses
        from below the mean to above the mean = 1 heartbeat
Step 5: BPM = (number of peaks / duration in seconds) × 60
Step 6: Buffer multiple readings → take the MEDIAN for stability
```

Key Parameters

Parameter	Value
Measurement Duration	20 seconds
Camera Used	Rear camera with flashlight
Valid BPM Range	45 – 180 BPM
Minimum Readings Required	3 readings for valid result
Stabilization Method	Median of last 15 buffered readings

Files Involved

File	Role
CameraScanScreen.kt	Camera UI, image analysis, raw signal collection

CameraScanViewModel1.kt	HR buffering, median calculation, state management
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Accuracy Notes

- This is a **real physiological measurement**, not an estimate
- Accuracy depends on: firm finger pressure, keeping finger still, adequate lighting
- Not medical-grade, but suitable for stress assessment purposes
- Same principle used by apps like Samsung Health, Google Fit, and Instant Heart Rate

2. Eye Blink Rate Measurement (Automatic Input)

How It Works

1. The **front camera** records a **15-second video** of the user's face
2. The video file is uploaded to the **Blink Detection API** (`blink-api-jxdv.onrender.com`)
3. The API uses a **machine learning model** to detect and count eye blinks in the video
4. The blink rate (blinks per minute) is returned and auto-filled in the assessment

Key Parameters

Parameter	Value
Recording Duration	15 seconds
Camera Used	Front camera
Processing	Server-side ML model (Blink API)
API Timeout	90 seconds (for cold-start servers)

Files Involved

File	Role
CameraScanScreen.kt	Video recording with progress bar
CameraScanViewModel.kt	Video upload to Blink API
Apis.kt	API endpoint definition (<code>/blink-count</code>)
ApiClient.kt	HTTP client with 90s timeout for video uploads

3. Sleep Duration (Automatic Input)

Data Source Priority (3-Layer Fallback)

Layer 1: Google Health Connect (Real Data)

Condition	Available?
Android 14+	✔ Health Connect is built-in (pre-installed)
Android 13 and below	✘ User must download "Health Connect" from Play Store

Important: Health Connect is a data hub — it does NOT track sleep by itself.
It requires a **source app** to provide sleep data, such as:

- Samsung Health
- Fitbit
- Sleep as Android
- A smartwatch (Galaxy Watch, Wear OS)

If Health Connect has sleep session data, the app reads the **longest sleep session** from the last 24 hours and returns the duration in hours.

Layer 2: Phone Inactive Time (UsageStats)

If Health Connect is unavailable or has no sleep data:

- Uses Android's **UsageStatsManager** to analyze app usage timestamps
- Finds the **longest gap** between consecutive phone usage events in the last 24 hours
- This gap represents when the phone was idle (likely while the user was sleeping)
- Example: If the user last used their phone at 11:30 PM and first opened it at 7:00 AM → sleep = 7.5 hours

Note: This requires the `PACKAGE_USAGE_STATS` permission and is a **proxy measurement**, not a direct sleep sensor.

Layer 3: Time-Based Estimate (Last Resort)

If both Layer 1 and Layer 2 fail:

Current Time	Estimated Sleep
5 AM – 11 AM (morning)	5 – 8 hours
12 PM – 11 PM (afternoon/evening)	7 hours
12 AM – 4 AM (late night)	4 hours

Note: This is a rough heuristic, not real data.

Summary Table

Layer	Source	Real Data?	Requires
1	Health Connect	✔ Yes	Android 14+ OR Health Connect app + fitness tracker
2	UsageStats (phone inactive time)	⚠ Proxy	PACKAGE_USAGE_STATS permission
3	Time-based estimate	✖ Estimate	Nothing

4. Steps Per Day (Automatic Input)

Data Source Priority (2-Layer Fallback)

Layer 1: Google Health Connect (Real Data)

Same requirements as sleep:

- Android 14+ has Health Connect built-in
- Needs a **source app** writing step data (Google Fit, Samsung Health, etc.)
- Reads all step records from **midnight today** to current time
- Returns the **total step count**

Layer 2: Time-Based Estimate (Fallback)

If Health Connect is unavailable or returns 0:

Time of Day	Estimated Steps
Before 7 AM	0 steps
7 AM	~350 steps
12 PM (noon)	~1,750 steps
3 PM	~2,800 steps
6 PM	~3,850 steps
10 PM	~5,250 steps

Formula: (hours since 7 AM) × 350 steps/hour

Note: Average person walks 5,000–8,000 steps/day.

Summary Table

Layer	Source	Real Data?	Requires
1	Health Connect	✓ Yes	Android 14+ OR Health Connect app + Google Fit/Samsung Health
2	Time-based estimate	✗ Estimate	Nothing

5. Android Version Compatibility

Android Version	Health Connect	Steps	Sleep	Heart Rate	Blink
14+ (API 34)	✓ Built-in	✓ Real (if Fit/Samsung Health active)	✓ Real (if tracked)	✓ Camera PPG	✓ Blink API
8–13 (API 26–33)	✗ Needs download	⚠ Estimate	⚠ Estimate / UsageStats	✓ Camera PPG	✓ Blink API

Key Takeaway

- **Heart Rate and Blink Rate** are always **real measurements** (camera + ML model)

- **Sleep and Steps** are real only if Health Connect is set up with a fitness tracker
 - The app **never crashes** regardless of device capabilities — it uses fallbacks gracefully
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6. Files Reference

File	Purpose
HealthConnectManager.kt	Sleep/Steps data with 3-layer fallback
CameraScanScreen.kt	Camera UI for blink recording and HR measurement
CameraScanViewModel.kt	HR buffering, blink video processing
AssessmentViewModel.kt	Assessment state, sensor data loading
AssessmentScreen.kt	6-step assessment UI
ApiClient.kt	HTTP clients with appropriate timeouts
Apis.kt	API endpoint definitions
AndroidManifest.xml	Permissions (Camera, Location, UsageStats)